NNN-T-580
July 16, 1963

SUPERSEDING
Int. Fed. Spec. GGG-T-00549a (GSA-FSS)
July 26, 1961
(See section 6)

FEDERAL SPECIFICATION

TONGS, LABORATORY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers laboratory tongs for the handling of laboratory apparatus such as beakers, flasks, crucibles, evaporating dishes, and casseroles, containing materials such as acids, caustics, and hot solutions.

1.2 Classification.

1.2.1 *Types, classes, compositions, and sizes.* The laboratory tongs covered by this specification shall be of the following types, and when applicable, the following classes, compositions, and sizes, as specified (see 6.2):

Type I.—Beaker tongs.
- Class 1.—Double-grip design (fig. 1).
- Class 2.—Weight-in-palm design (fig. 2).
- Class 3.—Craig design (fig. 3).

Type II.—Flask tongs.
- Class 1.—Double-grip design (fig. 4).
- Class 2.—Weight-in-palm design (fig. 5).

Type III.—Crucible tongs (fig. 6).
- Composition A. — Corrosion-resisting steel.
  - Size 1.—Short (9 to 10 inch length).
  - Size 2.—Long (18 to 20 inch length).
- Composition B. — Carbon steel, corrosion-resisting plated.
  - Size 1.—Short (9 to 10 inch length).
  - Size 2.—Long (18 to 20 inch length).
- Composition C.—Nickel-chromium alloy.
  - Size 1.—Short (9 to 10 inch length).
  - Size 2.—Long (18 to 20 inch length).
- Composition D.—High-purity nickel, low carbon.
  - Size 1.—Short (9 to 10 inch length).
  - Size 2.—Long (18 to 20 inch length).

Type IV.—Evaporating dish tongs (fig. 7).

Type V.—Casserole tongs (fig. 8).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

QQ-A-591—Aluminum Alloy Die Castings.
QQ-N-290—Nickel Plating (Electrodeposited).
QQ-N-301—Nickel; (for) Remelting.
QQ-P-416—Plating, Cadmium (Electrodeposited).
QQ-S-766 — Steel Plates, Sheets and Strip-Corrosion Resisting.
QQ-W-409—Wire, Steel, Carbon, (For Cold Heading and Cold Forging).
QQ-W-423—Wire, Steel, Corrosion-Resisting.

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UU-T-111 — Tape; Paper, Gummed (Sealing and Securing).
PPP-B-566 — Boxes, Folding, Paperboard.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-B-665—Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676—Boxes, Set-Up, Paperboard.
PPP-C-843—Cushioning Material, Cellulosic.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration *Regional Offices in Boston*, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-417 — Rubber Compositions, Vulcanized, General Purpose, Solid (Symbols and Tests).

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publication.** The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Classification Committee Publication:

Uniform Freight Classification Rules.

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 **Identification marking.** All tongs shall be marked in a legible and permanent manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.2 **Illustrations.** The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of tongs which are otherwise in accordance with the requirements of this specification.

3.3 **Type I.** Type I tongs shall be used for handling beakers.

3.3.1 *Class* 1. Type I, class 1, beaker tongs (fig. 1) shall be of the double-grip design. The handle shall consist of an upper and lower grip. The upper grip shall be stationary and shall have a raised, knurled, serrated, or similar top surface to facilitate gripping. The bottom of the lower grip shall be shaped for finger gripping. The lower grip shall pivot to actuate the jaws. The grips shall remain open under spring tension when not in use. In use, the tongs shall be actuated (closed) by squeezing the grips together. The jaws shall be 1 ± 1/8 inch in width and shall open and close to accommo-

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date beakers of 100 to 2000 milliliters capacity, inclusive (see 6.1.1). The jaws shall operate through a plane of a minimum of 3 inches below the handle. An adjusting screw with locknut on the front of the handle shall limit the span of the jaws. The jaws shall open to a maximum of 3-1/4 inches. The overall length of the tongs shall be not less than 13 inches. The jaws, fittings, and spring shall be of corrosion-resisting steel conforming to composition FS 302, condition B of QQ-W-423. The handgrips shall be of die-cast aluminum conforming to commercial alloy No. 43 of QQ-A-591. The jaws shall be covered with replaceable asbestos sleeves. The sleeves shall be non-raveling and woven from high quality asbestos yarn.

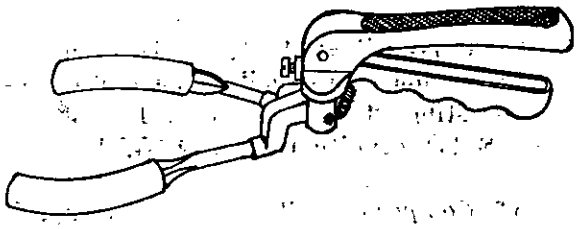


Figure 1.--Type 1, class 1, double-grip design beaker tongs.

3.3.2 *Class* 2. Type I, class 2, beaker tongs (fig. 2) shall be of the weight-in-palm design. The handle grips shall extend beyond the finger-holds so that the entire weight of the beaker (and contents) is supported by the palm of the hand. Finger-hold and thumb-hold shall be for right-hand gripping. The jaws shall be 1 ± 1/8 inch in width and shall open and close sufficiently to accommodate beakers of 100 to 1500 milliliters capacity, inclusive (see 6.1.2). The wires shall be flattened adjacent to the rivet to prevent shifting out of alignment. Rivets shall be spun to insure free action without binding. The overall length of the tongs shall be not less than 13 inches. The tongs shall be formed from corrosion-resisting steel wire, 1/4-inch nominal diameter, form I, composition FS 302, condition B of QQ-W-423. The jaws shall be covered with replaceable asbestos sleeves. The sleeves shall be non-raveling and woven from high quality asbestos yarn.

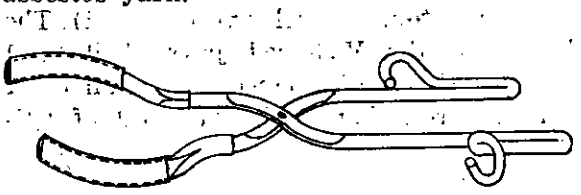


Figure 2.--Type I, class 2, weight-in-palm design beaker tongs.

3.3.3 *Class* 3. Type I, class 3, Craig design beaker tongs (fig. 3) shall have large finger-loops for hand gripping. The jaws shall open to at least 4-1/2 inches and accommodate beakers of 50 to 2000 milliliters capacity, inclusive (see 6.1.3). The wires shall be flattened adjacent to the rivet to prevent shifting out of alignment. Rivets shall be spun to insure free action without binding. The overall length of the tongs shall be not less than 9 inches. The tongs shall be formed from carbon steel wire, 1/4-inch nominal diameter, composition C1010, C1020, or C1025 of QQ-W-409 and corrosion-resisting plated with either cadmium, type I, class 1 of QQ-P-416, or nickel, class 2 of QQ-N-290. The jaws shall be covered with replaceable rubber sleeves; the rubber shall be type R, class optional, of MIL-STD-417.

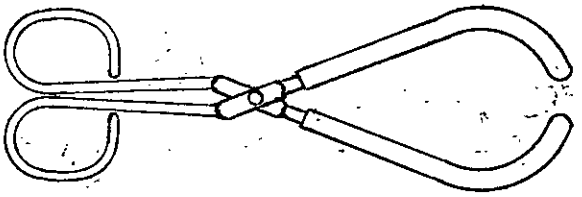


Figure 3.--Type I, class 3, Craig design beaker tongs.

3.4 **Type II.** Type II tongs shall be used for handling flasks.

3.4.1 *Class* 1. Type II, class 1, flask tongs (fig. 4) shall be of the double-grip design. The metals used, and the method of operation shall be the same as type I, class 1 tongs (see 3.3.1). The jaws shall open to at

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least 3-1/4 inches and accommodate various kinds of distillation flasks using No. 4 to No. 8 size stoppers, inclusive (see 6.1.4). The height of the V-shaped jaws shall be 1 ± 1/8 inch. The V-shaped jaws shall be of corrosion-resisting steel, class 302 of QQ-S-766.

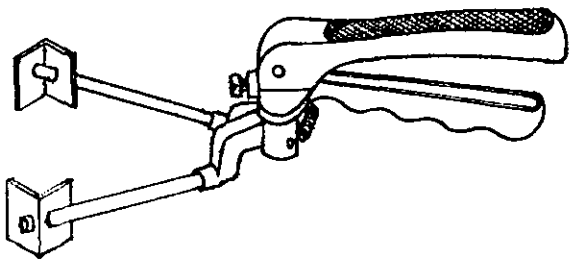


Figure 4.--Type II, class 1, double-grip design flask tongs.

3.4.2 *Class* 2. Type II, class 2, flask tongs (fig. 5) shall be of the weight-in-palm design. The metals used, and the method of operation shall be the same as type I, class 2 tongs (see 3.3.2). The jaws shall open to at least 3-1/4 inches and accommodate various kinds of distillation flasks using No. 4 to No. 8 stoppers, inclusive (see 6.1.5). The height of the V-shaped jaws shall be 1 ± 1/8 inch. The V-shaped jaws shall be of corrosion-resisting steel, class 302 of QQ-S-766.

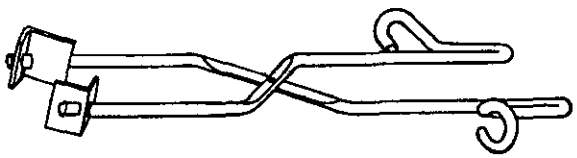


Figure 5.--Type II, class 2, weight-in-palm design flask tongs.

3.5 **Type III.** Type III crucible tongs (fig. 6) shall be of double-bent design and the gripping surfaces of the jaws shall be serrated (see 6.1.6). The two main wires shall be flattened adjacent to the rivet to prevent shifting out of alignment. Rivets shall be spun to insure free action without binding. The rivets shall be located 4-1/2 inches ± 1/4 inch from the tip of the jaws. Size 1 (short) crucible tongs shall have an overall length of 9 to 10 inches and shall be made from wire of 0.177-inch nominal diameter. Size 2 (long) crucible tongs shall have an overall length of 18 to 20 inches and shall be made from wire of 1/4-inch nominal diameter. The composition of the wire shall be in accordance with 3.5.1 to 3.5.4, as applicable.

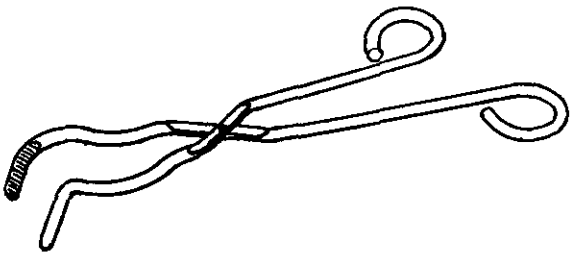


Figure 6.--Type III, crucible tongs.

3.5.1 *Composition A*. Type III, composition A crucible tongs shall be made of corrosion-resisting steel wire, form I, composition FS 430, condition A, of QQ-W-423.

3.5.2 *Composition B*. Type III, composition B crucible tongs shall be made of carbon steel wire, either composition C1010, C1020, or C1025 of QQ-W-409. The tongs shall be corrosion-resisting plated with either cadmium, type I, class 1 of QQ-P-416, or nickel, class 2 of QQ-N-290.

3.5.3 *Composition C*. Type III, composition C crucible tongs shall be made of nickel-chromium alloy, having 80 percent nickel and 20 percent chromium.

3.5.4 *Composition D*. Type III, composition D crucible tongs shall be made of high-purity, low carbon nickel, grade A of QQ-N-301.

3.6 **Type IV.** Type IV evaporating dish tongs (fig. 7) shall hold evaporating dishes having a top diameter from 75 millimeters (mm.) to 120 mm., inclusively (see 6.1.7) in such a manner that the dish cannot slip out of the tongs' grip. The jaws shall fit around the sides and grip the dish below its top edge. The end of each jaw shall have a

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prong projecting up and over the top edge of the dish. A third prong shall project from the hinge-joint up and over the top edge of the dish. The overall length of the tongs shall be not less than 9-1/2 inches. The tongs shall be made of corrosion-resisting steel wire 3/16-inch nominal diameter, form I, composition FS 302, condition B of QQ-W-423. The wires shall be flattened.

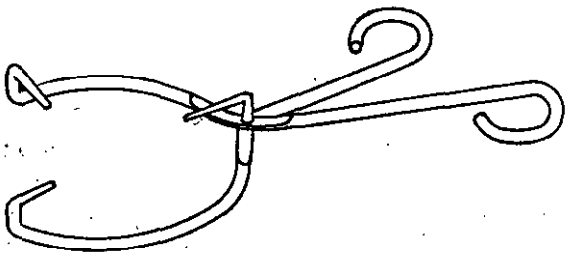


Figure 7.—Type IV, evaporating dish tongs.

3.7 **Type V.** Type V casserole tongs (fig. 8) shall grip the handle of the casserole dish in a firm and non-shifting manner (see 6.1.8). The lower jaw of the tongs shall be V-shaped to fit the handle of the casserole dish and the upper jaw shall firmly hold the handle of the casserole dish in the V-shaped lower jaw. The handle of the casserole tongs shall be of the weight-in-palm design (see 3.3.2). The wires shall be flattened adjacent to the rivet to prevent shifting out of alignment. Rivets shall be spun to insure free action without binding. The overall length of the tongs shall be not less than 9-1/2 inches. The tongs shall be made from corrosion-resisting steel wire, 3/16-inch nominal diameter, form I, composition FS 302, condition B of QQ-W-423. The V-shaped jaw shall be of corrosion-resisting steel, class 302 of QQ-S-766.

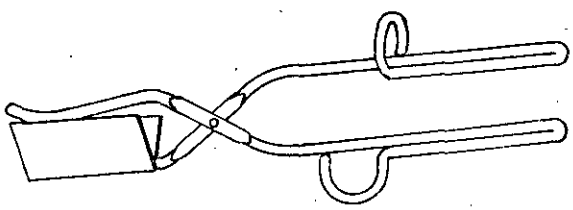


Figure 8.—Type V, casserole tongs.

3.8 **Workmanship.** Tongs shall be free from cracks, pits, scratches, grinding marks, sharp edges, or other defects or irregularities which impair their serviceability. The jaws shall be in proper alignment to insure a firm grip and the joint shall operate freely but firmly. Rivets shall be neatly and firmly set without binding the joint.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Supplier's responsibility.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection and test procedures.* The supplier may utilize any alternative inspection and test procedures which will insure equal or better quality by submitting a written proposal and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall furnish evidence acceptable to the contracting officer that the material requirements of section 3 and the packaging and packing requirements of section 5 have been satisfied.

4.3 **Lot.** A lot shall consist of those laboratory tongs of the same type, class, com-

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position and size produced by one manufacturer and submitted for delivery at one time.

4.4 **Sampling.** Sampling shall be conducted in accordance with MIL-STD-105.

4.5 **Examination.** Samples of tongs and level A and level B preparation for delivery requirements shall be examined in accordance with the classification of defects and MIL-STD-105.

4.5.1 *Classification of defects.*

4.5.1.1 *Tongs.*

Categories	Defects
Critical:	None defined.
Major:	*AQL 4.0 percent defective.*
101	Dimensions not as specified.
102	Jaws out of alignment or do not mate.
103	Jaws not serrated, when applicable.
104	Tongs bind or faulty rivet.
105	Sleeves not as specified.
106	Damaged (rust, cracks, pits, scratches, grinding marks, or sharp edges).
107	Identification marking illegible, incorrect or incomplete.

4.5.1.2 *Preparation for delivery (levels A and B).*

Categories	Defects
Critical:	None defined.
Major:	*AQL 2.5 percent defective.*
101	Unit, intermediate, or shipping container not as specified.
102	Number of tongs per unit or intermediate container not as specified.
103	Unlike tongs packaged per intermediate or shipping container.
104	Closure of unit, intermediate, or shipping container, not as specified.
105	Unit, intermediate, or shipping container damaged.
106	Cushioning material or blocking not as specified.
107	Marking of unit, intermediate, or shipping container illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packaging.** Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* Each tong shall be packaged in a box of appropriate size conforming to PPP-B-566, style I or II, type and class optional; PPP-B-636, type I, class 1; PPP-B-665, style optional; or PPP-B-676, type, class, and style optional. The box shall be provided with fiberboard blocking or sufficient cushioning material conforming to type II, class B of PPP-C-843 to prevent shifting of contents. Box closure shall be secured with tape conforming to UU-T-111.

5.1.1.2 *Intermediate packaging.* The number of unit packages specified (see 6.2) shall be intermediately packaged in a snug fitting box conforming to type I, class 1, of PPP-B-636. The box shall be closed in accordance with appendix of the box specification.

5.1.2 *Level C.* Tongs shall be packaged in accordance with the manufacturer's commercial practice suitable for shipment from the supply agency to the first receiving activity for immediate use.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Tongs shall be packed in a box conforming to PPP-B-621, class 2; or PPP-B-601, overseas type. As far as practical, exterior containers shall be of uniform size and shape, be of minimum cube and tare, and contain identical quantities of like items. Gross weight of each box shall not exceed approximately 200 pounds. Containers shall be closed and strapped in accordance with the applicable box specification or appendix thereto.

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5.2.2 *Level B.* Tongs shall be packed in accordance with 5.2.1 except that shipping containers shall conform to PPP-B-621, class 1, or PPP-B-601, domestic type.

5.2.3 *Level C.* Tongs shall be packed to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of carriers as applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.**

6.1.1 *Type I, class* 1. Type I, class 1, double-grip design beaker tongs are intended for handling hot beakers in a plane below the user's hand. The tongs are intended to accommodate beakers conforming to NNN-B-175, of 100 to 2000 milliliters capacity, inclusive (see 3.3.1).

6.1.2 *Type I, class* 2. Type I, class 2, weight-in-palm design beaker tongs are intended for handling hot beakers by supporting the beaker and its contents with the palm of the hand. The tongs are intended to accommodate beakers conforming to NNN-B-175, of 100 to 1500 milliliters capacity, inclusive (see 3.3.2).

6.1.3 *Type I, class* 3. Type I, class 3, Craig design beaker tongs are intended for general purpose use. The tongs are intended to accommodate beakers conforming to NNN-B-175, of 50 to 2000 milliliters capacity, inclusive (see 3.3.3).

6.1.4 *Type II, class* 1. Type II, class 1, double-grip design flask tongs are intended to handle hot distillation flasks with the flask contents in a plane below the user's hand. The tongs are intended to handle flasks using No. 4 to No. 8 stoppers, inclusive (see 3.4.1).

6.1.5 *Type II, class* 2. Type II, class 2, weight-in-palm design flask tongs are intended to handle hot distillation flasks by supporting the flask and its contents with the palm of the hand. The tongs are intended to handle flasks using No. 4 to No. 8 stoppers, inclusive (see 3.4.2).

6.1.6 *Type III.* Type III crucible tongs are intended for general laboratory use (see 3.5). The size 1 (short) tongs are used for handling hot crucibles and similar operations. The size 2 (long) tongs are intended for handling crucibles in and out of muffle furnaces. The various material compositions of type III tongs provide for handling crucibles containing laboratory solutions of different degrees of corrosiveness. Composition C tongs are intended for handling platinum ware.

6.1.7 *Type IV.* Type IV evaporating dish tongs are intended to accommodate evaporating dishes conforming to NNN-D-340, of 70 to 250 milliliters capacity, inclusive of type I, and types II and III (see 3.6).

6.1.8 *Type V.* Type V casserole tongs are intended to accommodate casseroles conforming to types I and II of NNN-C-25 (see 3.7).

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.

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(*b*) Type; and when applicable, class, composition, and size required (see 1.2.1).

(*c*) Levels of packaging and packing required (see 5.1 and 5.2).

(*d*) Number of tongs required per intermediate container (see 5.1.1.2).

6.3 Military use. Tongs procured under this specification for military use are limited to those shown on the applicable military standards. Personnel of the military agencies should refer to these documents for guidance.

6.4 Supersession data. This specification includes the requirements of Interim Federal Specification GGG-T-549(GSA-FSS), dated February 6, 1952 and Military Specification MIL-T-996(BuMed), dated October 15, 1949. The crucible tongs of the superseded specifications are covered in this specification as shown below:

6.5 Transporation description. Transportation description applicable to this item is:

Tools, not otherwise indexed by name. Carload minimum weight 30,000 pounds.

Tools, hand, not otherwise indexed. Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—MU

Navy—Md

Air Force

OTHER INTERESTS:

Army—E M O

Superseded specification				Replacement tongs per GGG-T-00549a (GSA-FSS) and NNN-T-580		
Specification	Type	Class	Size	Type	Composition	Size
GGG-T-549 (GSA-FSS)	I	A	8	III	A	1
	I	A	9	III	A	1
	I	A	10	III	A	1
	I	B	8	III	B	1
	I	B	9	III	B	1
	I	B	10	III	B	1
	I	C	8	III	A, B, or D	1
	I	C	9	III	A, B, or D	1
	I	C	10	III	A, B, or D	1
	II	—	8	III	C	1
	II	—	9	III	C	1
	II	—	10	III	C	1
MIL-T-996 (BuMed)	—	—	—	III	A	1

Copies of this specification may be purchased for 5 cents each.

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